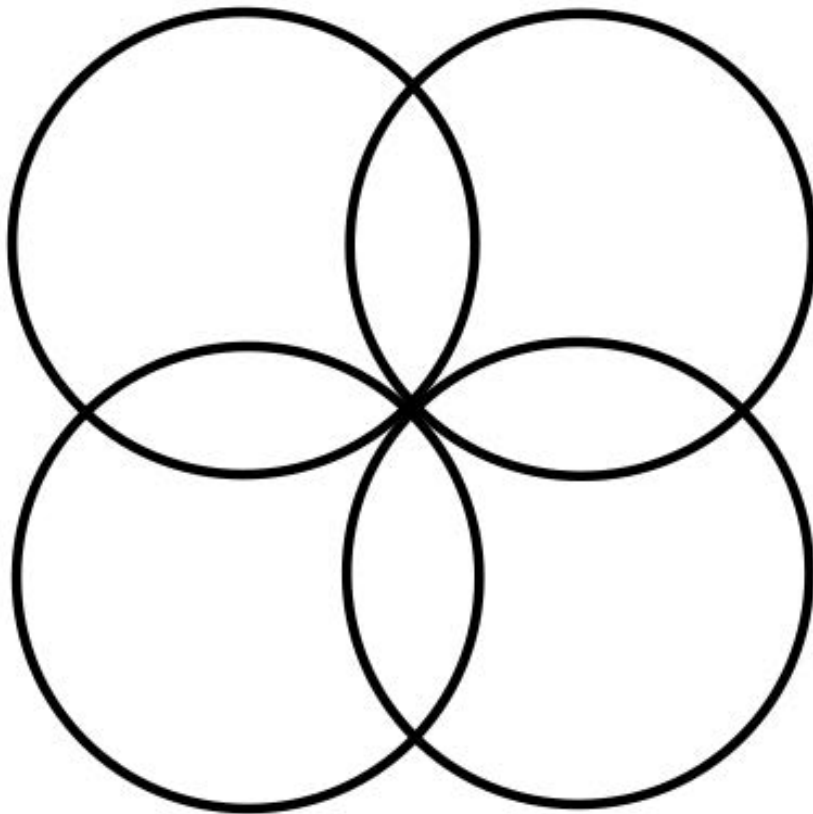


Puzzle of the Week

Equal Sums – 4

THE CHALLENGE: Here is a diagram created by overlapping four circles. The overlapping circles create eight regions. Put a number in each of the eight regions, using each of the numbers 1 to 8 exactly once, so that the sum of the numbers in each circle is the same.



1 2 3 4 5 6 7 8

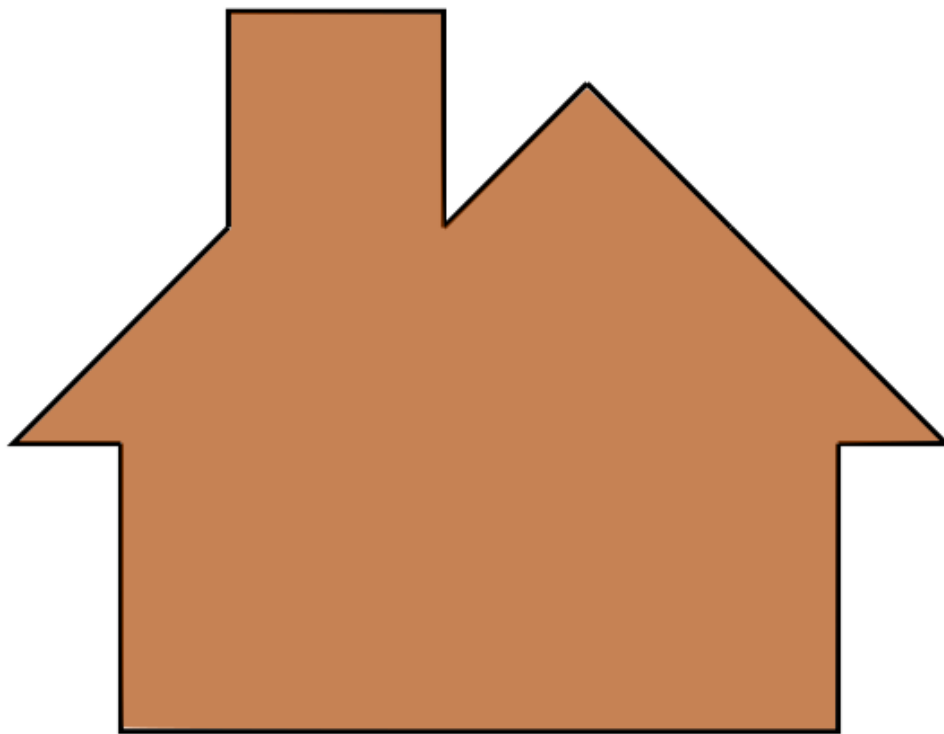
EXPLORATION: How many different answers can you find? Do you think there are any more? What happens if you use other number ranges? Are there other interesting problems like this with intersecting circles?

Puzzle of the Week

Finding the Pieces – 3

A **trapezoid** is a four-sided figure that has exactly one pair of parallel sides (parallel lines in a surface are lines that never meet). A **parallelogram** is a four-sided figure that has two pairs of parallel sides. A **rectangle** is a four-sided figure with four right angles. A **square** is a rectangle with four equal sides. A **right triangle** is a triangle with a right angle.

THE CHALLENGE: Break this figure up using trapezoids, parallelograms, rectangles, squares and right triangles. Use as few pieces as you can.

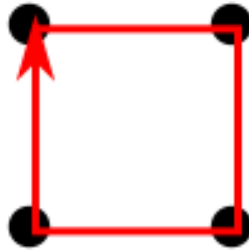


EXPLORATION: How many ways can you find to do this using this fewest number of pieces?

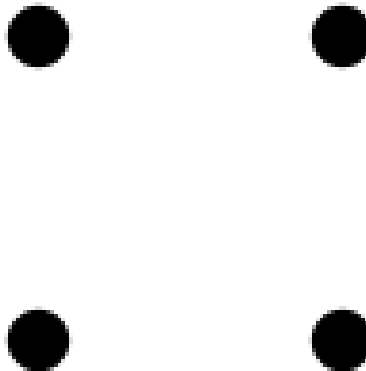
Puzzle of the Week

Lines – 1

Here is a 2 by 2 array of dots. Four connected line segments have been drawn in a path that begins and ends at the same point, and that goes through all four points.



THE CHALLENGE: Find **three** connected line segments that create a path that begins and ends at the same point, and that goes through all four points.

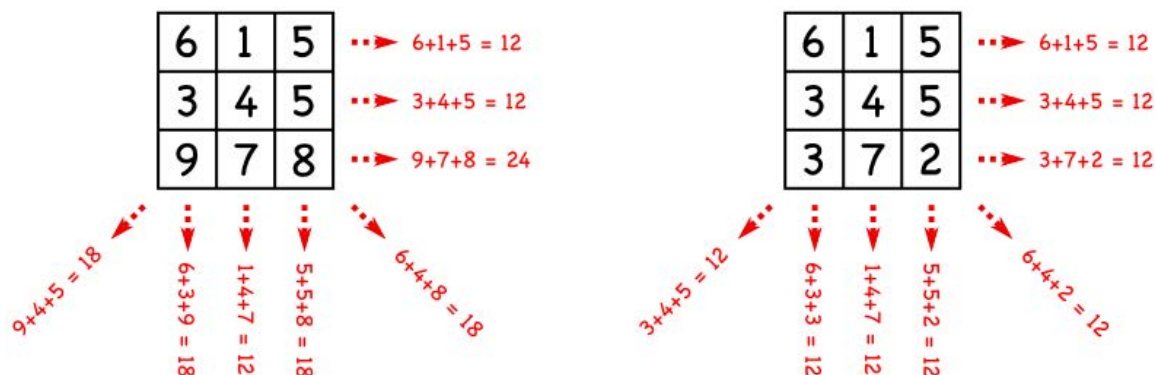


EXPLORATION: Play around with other grids and discover shortest paths that go through all the points.

Puzzle of the Week

Magic Squares – 4

In a **Magic Square**, all the rows, columns and diagonals add up to the same number. This first square is not a Magic Square. The second one is a Magic Square with a constant sum of 12.



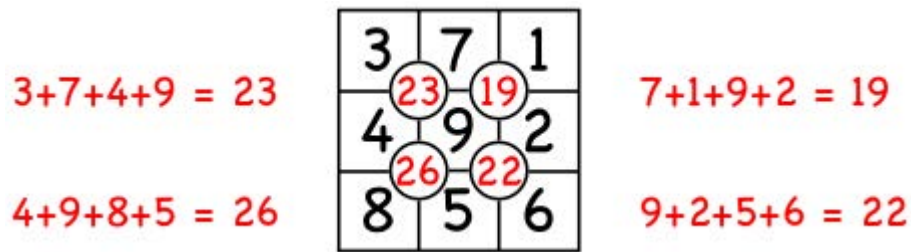
THE CHALLENGE: Fill in these two Magic Squares using these two sets of numbers: 1) {6, 7, 8, 9, 10, 11, 12, 13, 14} and {3, 6, 9, 12, 15, 18, 21, 24, 27}.

EXPLORATION: Look at all the Magic Squares you've seen that don't have duplicate entries. Make a table of your results and look for patterns. If you were given any other sequence of numbers that are evenly spaced, such as {4, 9, 14, 19, 24, 29, 34, 39, 44}, would you be able to immediately fill in the Magic Square?

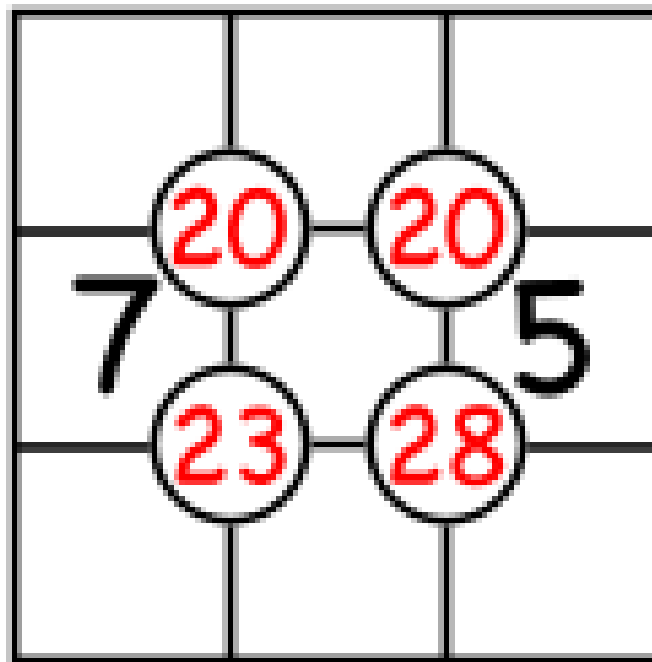
Puzzle of the Week

Sujiko Puzzle – 3

In a Sujiko puzzle, use each of the numbers from 1 to 9 once in the nine squares. The number in each circle must be the sum of the four squares that surround it.



THE CHALLENGE: Fill in this Sujiko puzzle.



1 2 3 4 6 8 9

Puzzle of the Week

Letter Substitutions – 4

Rules:

1. A letter represents a digit from 0 to 9, and has the same value throughout a single puzzle.
2. No number can start with the digit 0.
3. Within a puzzle, different letters must have different values.

$$\begin{array}{r}
 8 \\
 + A \\
 \hline
 B \ 2
 \end{array}
 \Rightarrow
 \begin{array}{r}
 8 \\
 + 4 \\
 \hline
 1 \ 2
 \end{array}$$

THE CHALLENGE: Find the value of A, B, C, D, and E in these puzzles.

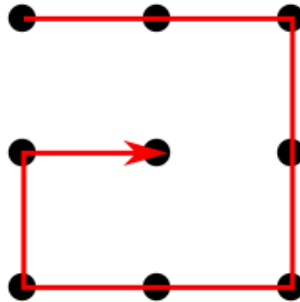
$$\begin{array}{r}
 A \\
 + B \ B \\
 \hline
 C \ 7
 \end{array}
 \qquad
 \begin{array}{r}
 D \\
 + E \ D \\
 \hline
 D \ E
 \end{array}$$

EXPLORATION: Make some letter substitution puzzles for your friends to solve.

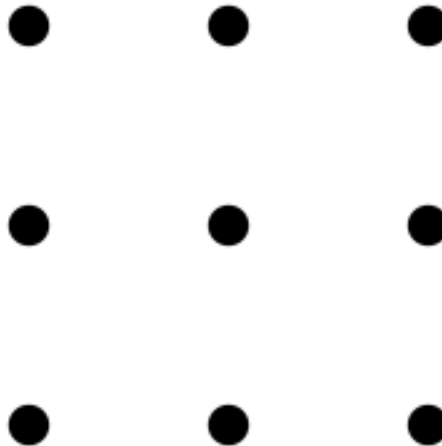
Puzzle of the Week

Lines – 2

Here is a 3 by 3 array of dots. Five connected line segments have been drawn in a path that goes through all nine points.



THE CHALLENGE: Find **four** connected line segments that create a path that goes through all nine points.



EXPLORATION: Play around with other grids and discover shortest paths that go through all the points.

Puzzle of the Week

Odd Squares

An *Odd Square* is a square grid of numbers in which the numbers in each row and column add up to an odd number. This is a 3 by 3 Odd Square using the numbers from 3 to 11.

9	3	7	→ $9+3+7 = 19$
6	4	5	→ $6+4+5 = 15$
8	10	11	→ $8+10+11 = 29$

↓ $9+6+8 = 23$	↓ $3+4+10 = 17$	↓ $7+5+11 = 23$
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THE CHALLENGE: Use the numbers from 1 to 9 to form a 3 by 3 Odd Square.

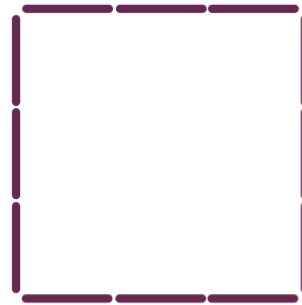
1 2 3 4 5 6 7 8 9

EXPLORATION: Define what an Even Square would be and consider the Even Squares that can be formed by the numbers from 1 to 9.

Puzzle of the Week

Stick Areas

Suppose you have 12 sticks to use to enclose a region with horizontal and vertical sides. These three possible ways for doing that enclose areas of size 3, 5, and 9.



THE CHALLENGE: Find all the possible areas that you can create with 12 sticks.

EXPLORATION: What happens when you change the number of sticks? Are there some numbers of sticks that don't work at all?

Puzzle of the Week

Letter Substitutions – 5

Rules:

1. A letter represents a digit from 0 to 9, and has the same value throughout a single puzzle.
2. No number can start with the digit 0.
3. Within a puzzle, different letters must have different values.

$$\begin{array}{r}
 8 \\
 + A \\
 \hline
 B \ 2
 \end{array}
 \Rightarrow
 \begin{array}{r}
 8 \\
 + 4 \\
 \hline
 1 \ 2
 \end{array}$$

THE CHALLENGE: Find the value of B, E, S, T, O, G, and U in these puzzles.

$$\begin{array}{r}
 B \ E \\
 + B \ E \\
 \hline
 S \ E \ E
 \end{array}
 \qquad
 \begin{array}{r}
 T \ O \\
 + G \ O \\
 \hline
 O \ U \ T
 \end{array}$$

EXPLORATION: Make some letter substitution puzzles for your friends to solve.

Puzzle of the Week

Fill in the Blanks – 6

These sums, using the numbers from 1 to 6 once each, are not particularly close to 100.

$$\begin{array}{r}
 \boxed{3} \ \boxed{1} \\
 \boxed{4} \ \boxed{2} \\
 + \ \boxed{6} \ \boxed{5} \\
 \hline
 1 \quad 3 \quad 8
 \end{array}$$

$$\begin{array}{r}
 \boxed{1} \ \boxed{3} \\
 \boxed{2} \ \boxed{5} \\
 + \ \boxed{4} \ \boxed{6} \\
 \hline
 8 \quad 4
 \end{array}$$

THE CHALLENGE: Use the numbers from 1 to 6 once each to make a sum as close to 100 as possible.

$$\begin{array}{r}
 \boxed{} \ \boxed{} \\
 \boxed{} \ \boxed{} \\
 + \ \boxed{} \ \boxed{} \\
 \hline
 \end{array}$$

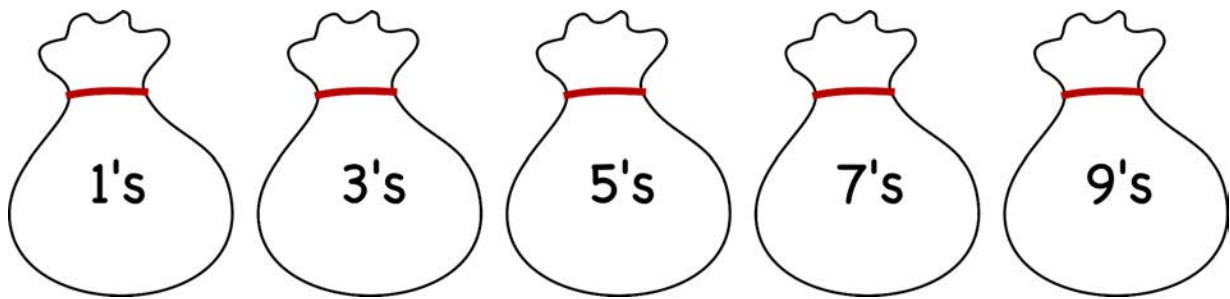
1 2 3 4 5 6

EXPLORATION: How does your answer change if you use the numbers from 1 to 7 or 1 to 8 instead, using each number no more than once.

Puzzle of the Week

Adding Ten Numbers

THE CHALLENGE: You have five big bags of coins. Each bag has only one kind of coin. The bags have coins worth 1, 3, 5, 7, and 9. If you can, find ten coins that add up to 43. If you can't, describe why it is impossible.

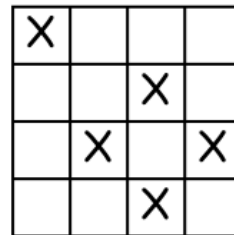
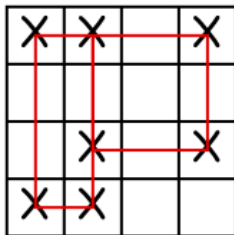


EXPLORATION: What are all the possible numbers you can make with ten coins from these bags?

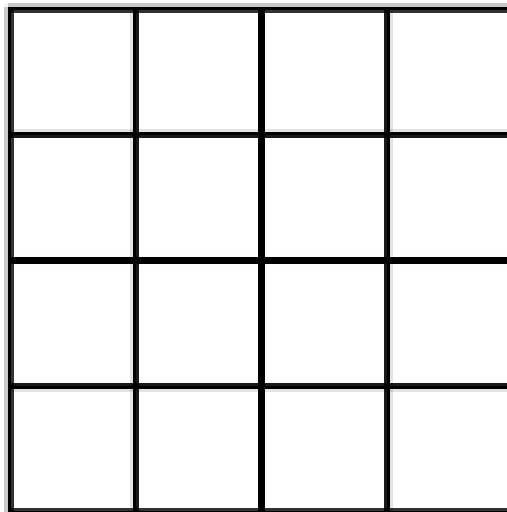
Puzzle of the Week

Avoiding Rectangles – 2

The X's in a grid can become the corners of rectangles with horizontal and vertical sides. The X's in this first grid form two rectangles. However, the goal is to avoid creating rectangles. The X's in the second grid are placed to avoid forming any rectangles.



THE CHALLENGE: Place as many X's as you can in this 4 by 4 grid and avoid creating any rectangles.

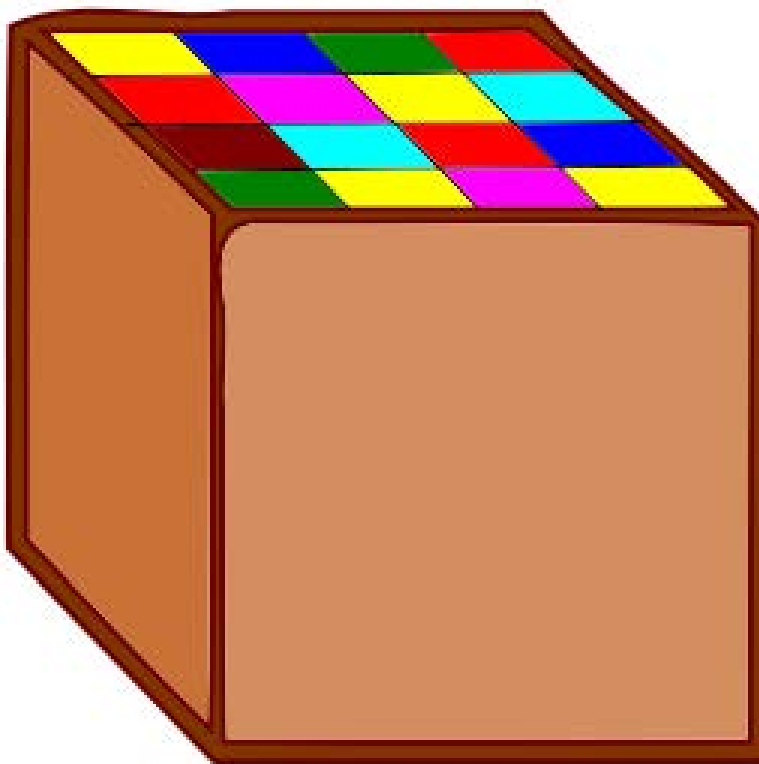


EXPLORATION: What is the most you can have for other sizes of grids? Do you see any patterns in your answers?

Puzzle of the Week

Boxed Blocks

THE CHALLENGE: There is a wooden box, without a lid, holding a $4 \times 4 \times 4$ collection of 64 blocks. How many of the blocks touch some part of the box?

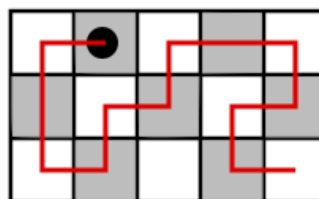
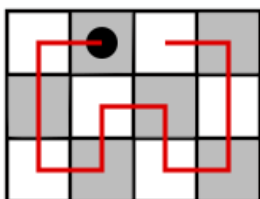


EXPLORATION: What happens for a $5 \times 5 \times 5$ collection of 125 blocks in a bigger box? What about other sizes of boxes? How do these answers change if the boxes have a lid that touches the top row of blocks?

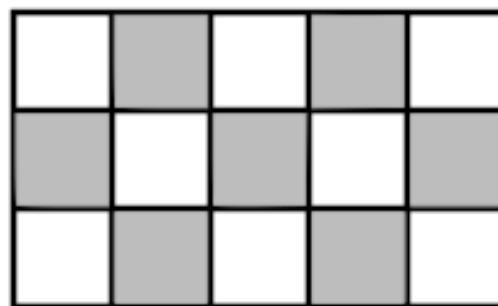
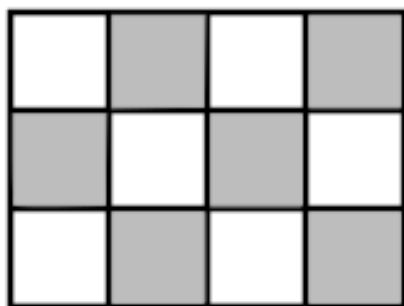
Puzzle of the Week

Paths on Checkerboards

The first 3 by 4 checkerboard has a path that visits every square starting at the black dot. The second 3 by 5 checkerboard has no path that visits every square starting at the black dot.



THE CHALLENGE: For these two checkerboards, identify which starting positions will begin paths that visit every square in the board and which ones do not. What is the difference?

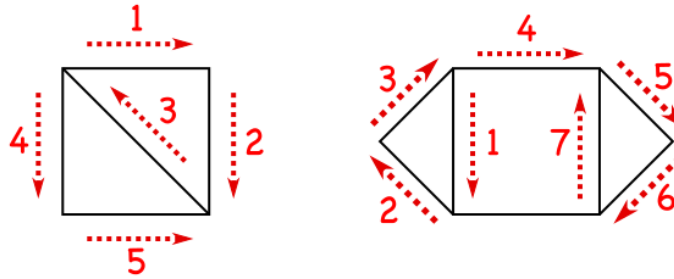


EXPLORATION: Create some other sizes of checkerboards and try various starting positions on these. Do you see any patterns for which starting positions work on each board?

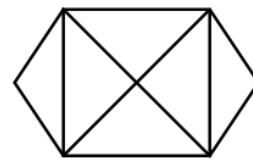
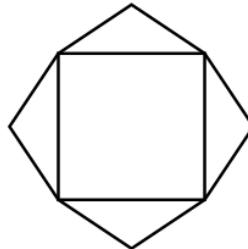
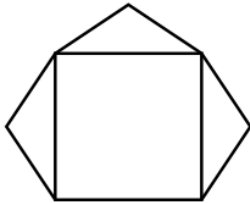
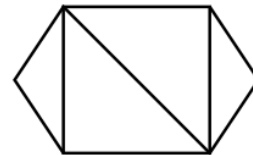
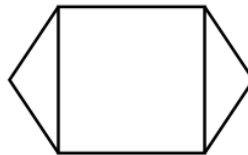
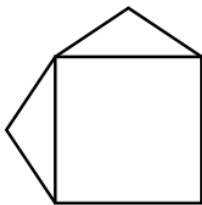
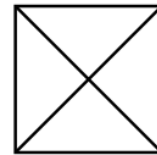
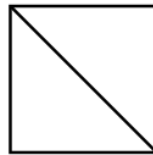
Puzzle of the Week

Parades – 1

Parades wish to visit each street on their route exactly once. In these two examples, the first one is a successful parade route and the second one is not (one street is left out).



THE CHALLENGE: For each street layout, either find a parade route that visits each street exactly once or decide that it is impossible. For street layouts that have a parade, which ones allow parades to start and end at the same place? Can you find a pattern in your results?



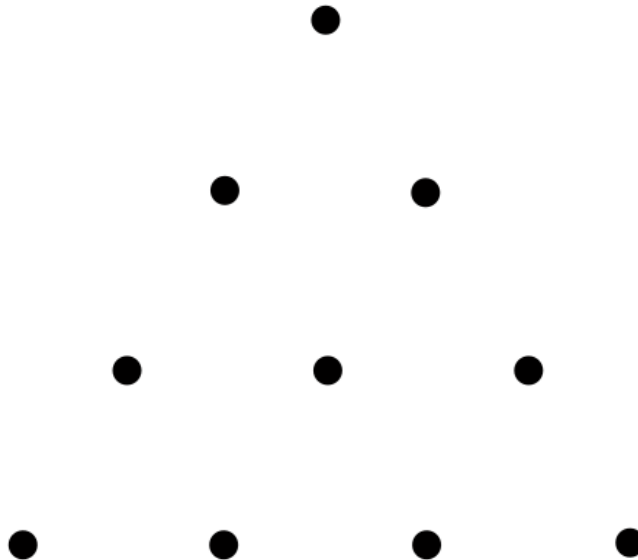
Puzzle of the Week

Avoiding Triangles

A *regular triangle* is a triangle with equal sides and equal angles. In this first pyramid, two regular triangles have been made using the dots. In the second pyramid, enough dots have been crossed out that it is impossible to find three dots that form a regular triangle.



THE CHALLENGE: What is the fewest number of dots you can remove from this pyramid so that there are no equilateral triangles of any size or orientation formed by the dots?



EXPLORATION: Can you cross out fewer dots than the introductory example to solve the six-dot pyramid? Do you see a pattern of solutions that would suggest solutions for larger pyramids?

Puzzle of the Week

Removing Digits

THE CHALLENGE: Which ten digits would you remove (they don't need to be next to each other) from the number 12345123451234512345 to make the new number as large as possible? Which ten digits would you remove to make it as small as possible?

1234512345123451234512345

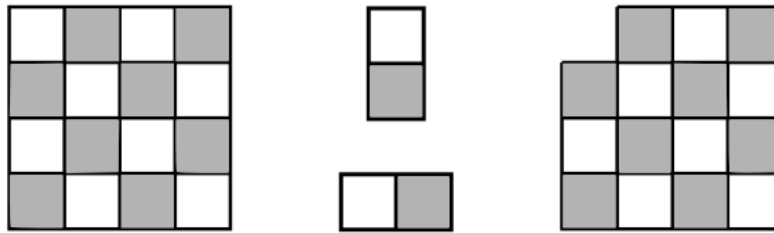
EXPLORATION: Can you think of other interesting problems similar to this one?



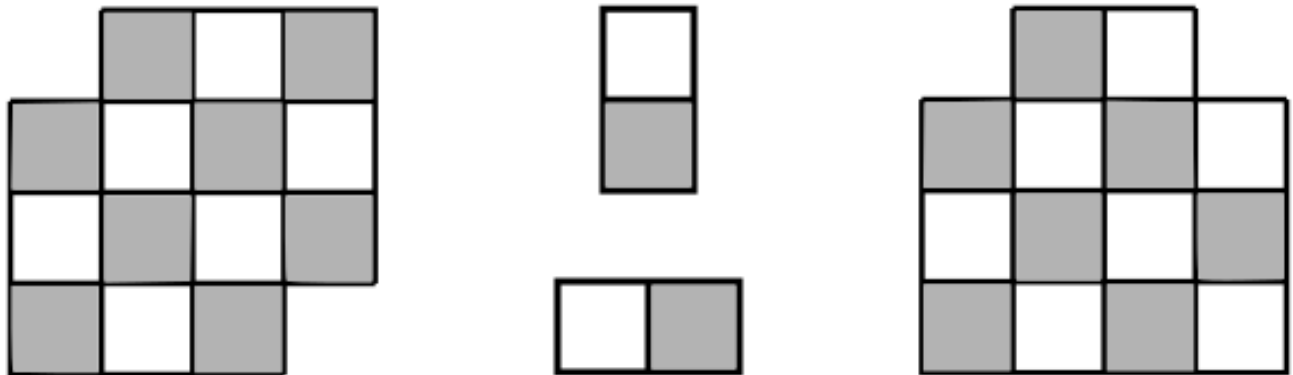
Puzzle of the Week

Dominoes on Checkerboards

The first of these two checkerboards is easy to cover with dominoes. The second one, which has one square missing, is impossible to cover with dominoes.



THE CHALLENGE: Describe why one of these checkerboards is easy to cover with dominoes and the other one is impossible.

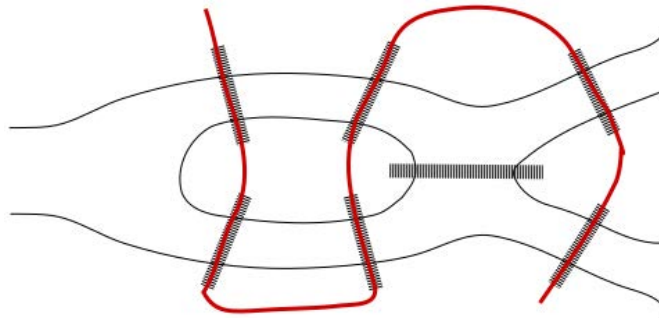


EXPLORATION: Explore which pairs of small squares you can remove from the original board and still be able to exactly cover the remaining board with dominoes.

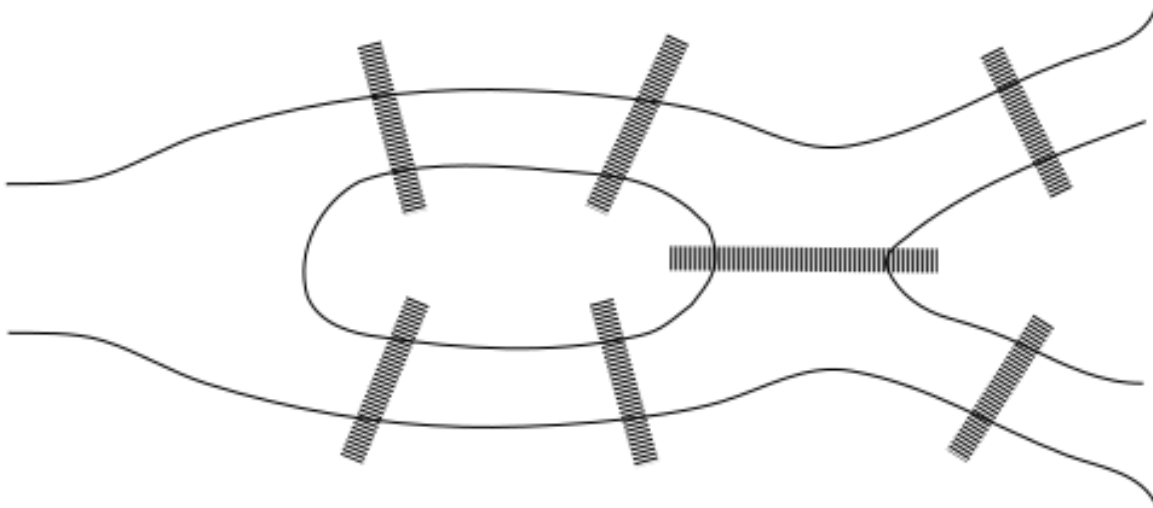
Puzzle of the Week

Parades – 2 – 7 Bridges of Königsberg

This is a map of Königsberg showing the river that runs through it, the island in the middle of the river, and the seven bridges that span the river. The people of Königsberg wanted to have a parade that went over each of their bridges exactly once. They could not seem to find a parade route that visited all seven bridges.



THE CHALLENGE: If you can, find a parade route that crosses each of their bridges exactly once. If you can't, give a reason for why it is impossible.



EXPLORATION: If you add another bridge over the river, would that make the problem easier or harder?

Puzzle of the Week

Parity – 3 – Last Number Standing

The numbers 1 to 5 are written on a board. Then pairs of numbers are selected, erased, and replaced by their difference. This continues until there is a single number remaining. In the example below, that number is 1.

$$1 \ 2 \ 3 \ 4 \ 5 \Rightarrow 1 \ 2 \ 4 \ 2$$

$$1 \ 2 \ 4 \ 2 \Rightarrow 2 \ 2 \ 3$$

$$2 \ 2 \ 3 \Rightarrow 2 \ 1$$

$$2 \ 1 \Rightarrow 1$$

THE CHALLENGE: How small can that single number be? Can it be 0? Does your answer change if the numbers go from 1 to 6, or from 1 to 7?

$$1 \ 2 \ 3 \ 4 \ 5 \Rightarrow ?$$

EXPLORATION: For a given list of numbers, such as 1 to 5, which last numbers are possible? What is the smallest and largest possible last number? When can 0 or the top number occur on the list of possibilities?

Puzzle of the Week

Product Equals Sum

THE CHALLENGE: Suppose you have five positive whole numbers, A, B, C, D, and E, so that the following is true. What is the largest possible value of any of these numbers?

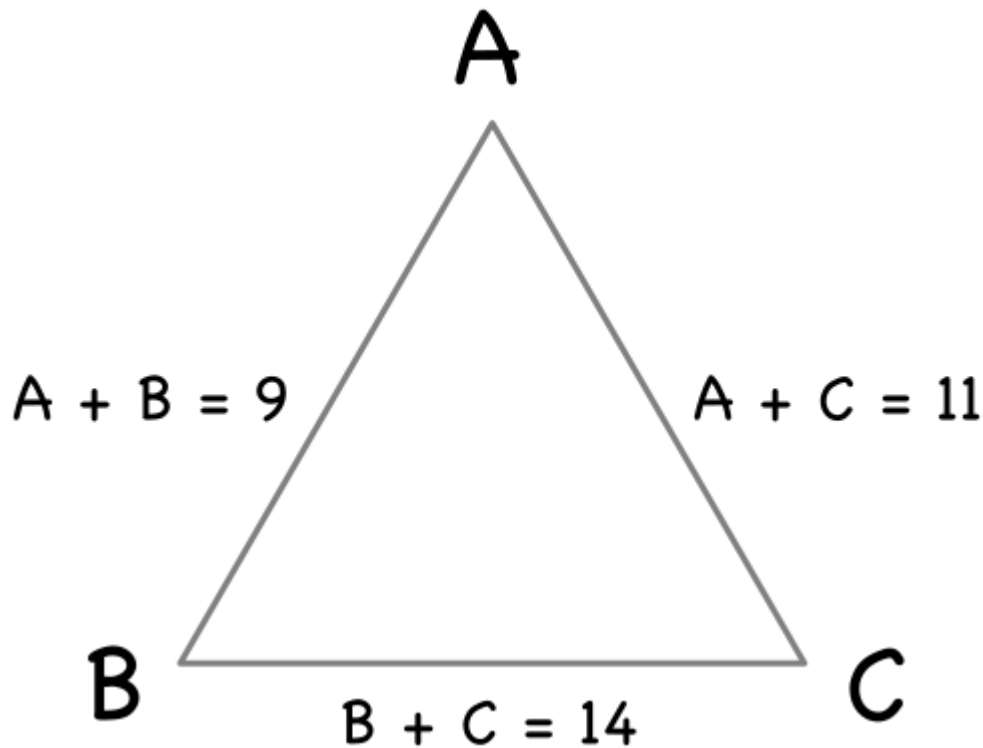
$$A \times B \times C \times D \times E = A + B + C + D + E$$

EXPLORATION: What would happen if you had a different number of numbers in this kind of equation?

Puzzle of the Week

Mystery Sums – 1

THE CHALLENGE: There are secret numbers at the corners of a triangle. The sums of these numbers are written in the middle of each side. The numbers in the middle of the sides are: 9, 11, and 14. Find the secret numbers.



EXPLORATION: Make up a few of these problems for your friends. Can you come up with a general method for solving this kind of puzzle? Can you solve these for polygons with more sides?

Puzzle of the Week

Fractions – 1

THE CHALLENGE: Use the numbers 1 to 9 at most once each to fill in the boxes. Can you find more than one solution?

$$\begin{array}{c} \square \\ \hline \square \end{array} = \begin{array}{c} \square \\ \hline \square \end{array} = \begin{array}{c} \square \\ \hline \square \square \end{array}$$

1 2 3 4 5 6 7 8 9

Puzzle of the Week

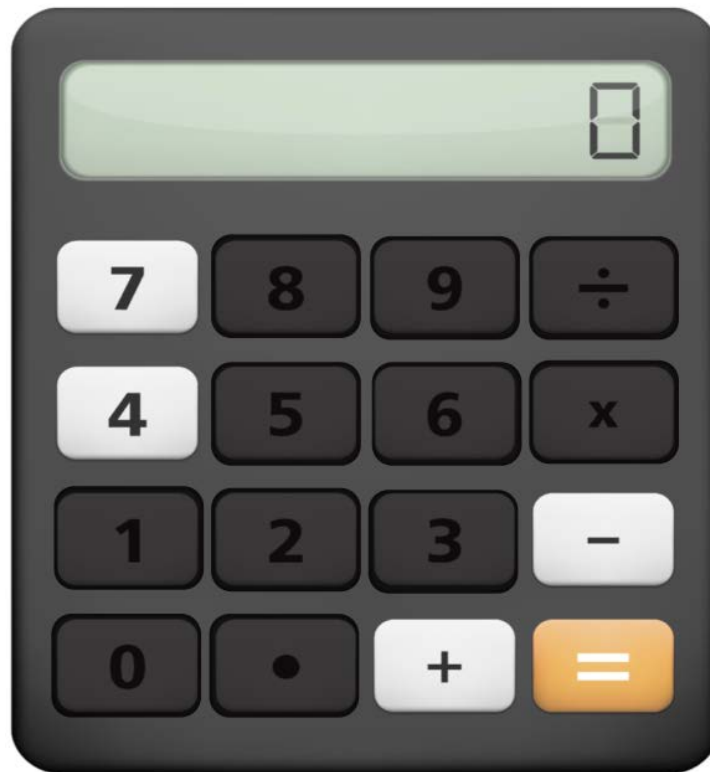
Broken Calculator – 1

You have a calculator that is badly broken. Its only working keys are 4, 7, +, and -. Even with this limited ability, it is possible to make every number. For example:

$$1 = 4 + 4 - 7$$

$$2 = 4 + 4 + 4 + 4 - 7 - 7$$

THE CHALLENGE: Show that this calculator can make all the numbers from 1 to 12.



EXPLORATION: Replace 4 and 7 by other pairs of numbers. When is it possible to get all the numbers from 1 to 12, and when is it impossible? Can you look at a pair of numbers and predict what will happen?

Puzzle of the Week

Maximizing Products with 16

Look at ways to break up 16 into a sum of numbers that you can then multiply to get as big a product as possible. Writing $16 = 10 + 6$ is a start, but $16 = 6 + 5 + 5$ is better. Can you do better?

$$16 = 10 + 6 \text{ and } 10 \times 6 = 60$$

$$16 = 6 + 5 + 5 \text{ and } 6 \times 5 \times 5 = 150$$

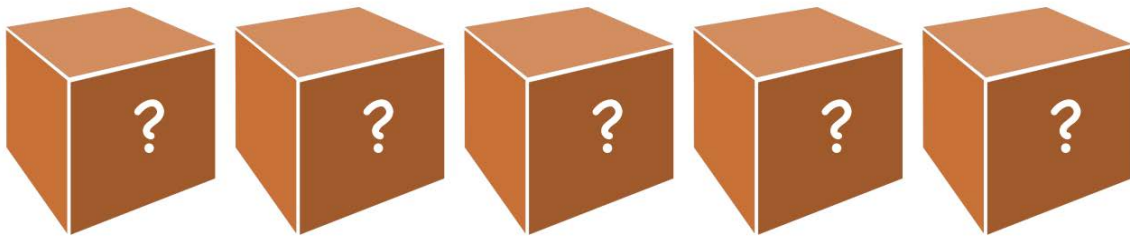
THE CHALLENGE: What is the biggest product you can make by breaking 16 into a sum of numbers?

EXPLORATION: How does your strategy change if you replace 16 with 20, 50, or 100?

Puzzle of the Week

Mystery Sums – 2

THE CHALLENGE: There are five boxes to weigh and mail, and each of them weighs under 20 pounds. Unfortunately, the one available scale only weighs things over 20 pounds. The packages, weighed in pairs, weigh 22, 24, 25, 26, 27, 28, 29, 30, 32, and 33 pounds. How much does each package weigh?



22 24 25 26 27 28 29 30 32 33

EXPLORATION: Make up a few of these problems for your friends. See if you can come up with a general method for solving this kind of puzzle.

Puzzle of the Week

Fractions – 2

THE CHALLENGE: Use the numbers 1 to 9 at most once each to fill in the boxes. Make $\frac{\square}{\square} + \frac{\square}{\square}$ as small as possible. Also, use the numbers 1 to 9 at most once each to make $\frac{\square}{\square} + \frac{\square}{\square}$ as large as possible.

$$\frac{\square}{\square} + \frac{\square}{\square}$$

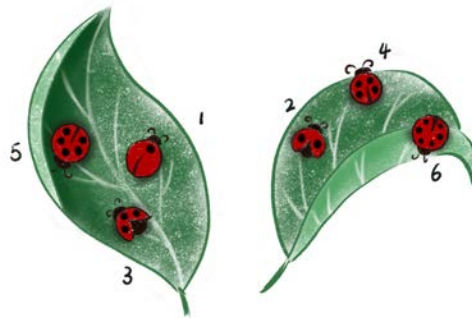
1 2 3 4 5 6 7 8 9

EXPLORATION: What happens if the fractions must be proper fractions?

Puzzle of the Week

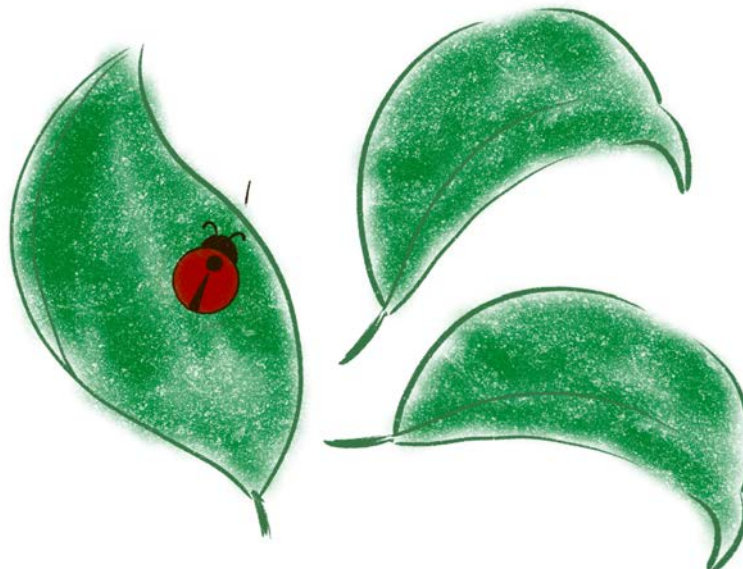
Ladybugs that don't Add Up – 2

Numbered ladybugs are landing on two leaves. The rule is: the number of dots of two ladybugs on a leaf cannot add up to the number of dots on another ladybug on that leaf. The leaf on the left is fine, but the leaf on the right has $2 + 4 = 6$.



THE CHALLENGE: Starting at 1 and counting up, how high can you go putting the numbered ladybugs on three leaves while following the rule for each of the leaves.

EXPLORATION: How do things change if you use only even numbers? How do things change if you use only odd numbers? What happens if you use more than three leaves?



Puzzle of the Week

Equal Products

THE CHALLENGE Find seven different single digits that make these three products the same.

$$\boxed{A} \quad B \quad \boxed{C} \quad D \quad \boxed{E} \quad F \quad G$$

$$A \times B \times C = C \times D \times E = E \times F \times G$$

1 2 3 4 5 6 7 8 9

EXPLORATION: After you solve this, think about what makes this an interesting puzzle. How might you change the puzzle and still keep it interesting? What happens if you use addition instead of multiplication?

Puzzle of the Week

Combining Digits – 1 2 4 8

Here are some ways to get 0 and 1 using 1, 2, 4, and 8.

$$0 = 8 - 1 * 2 * 4$$

$$0 = 8 * 1 - 2 * 4$$

$$1 = 8 - 2 * 4 + 1$$

$$1 = 8 - 4 - 2 - 1$$

THE CHALLENGE: How many numbers can you get using each of the numbers 1, 2, 4, and 8 in any order, using addition, subtraction, and multiplication?

EXPLORATION: What happens with other groups of four numbers? What happens if you use the five numbers: 1, 2, 4, 8, and 16?

Puzzle of the Week

Fractions – 3

THE CHALLENGE: Use the numbers 1 to 9 at most once each to fill in the boxes. Find two mixed numbers, whose fractional parts are proper fractions, that have as small a difference as possible.

$$\boxed{} \frac{\boxed{}}{\boxed{}} - \boxed{} \frac{\boxed{}}{\boxed{}}$$

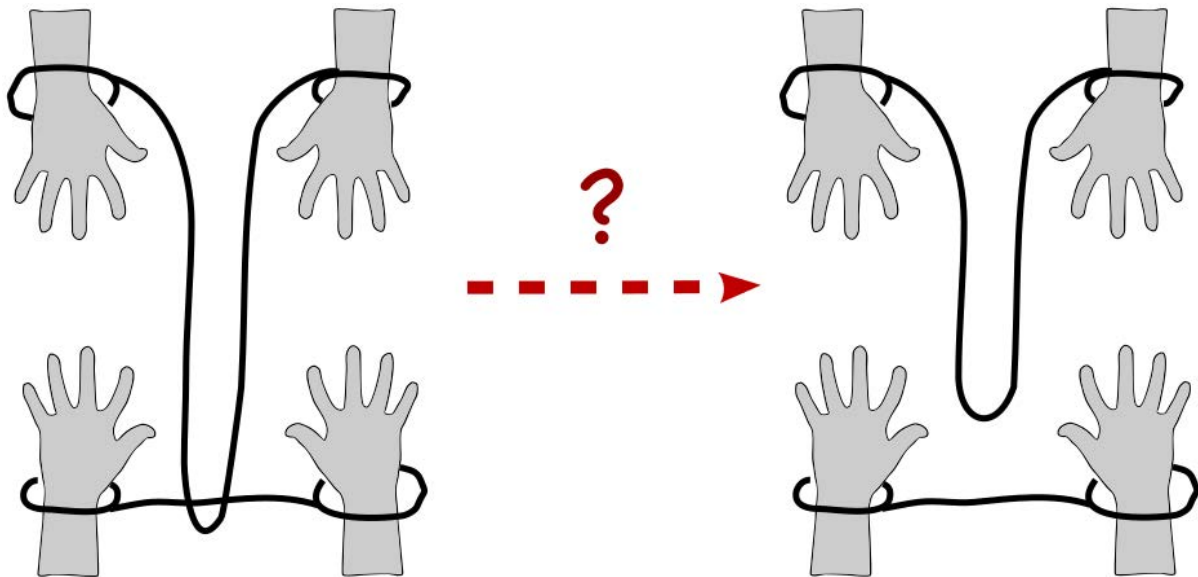
1 2 3 4 5 6 7 8 9

EXPLORATION: How does your answer change if you allow the fractional parts to be improper?

Puzzle of the Week

Are Their Hands Tied?

THE CHALLENGE: There are two people. Each person ties their hands together (loosely) with a string. As this is being done, the strings loop inside each other as shown. How do the two people get apart without untying their strings?



Puzzle of the Week

Extreme Products – 1

THE CHALLENGE: 1) Using the digits from 1 to 9, each at most once, make two 2-digit numbers whose product is as large as possible. 2) Also, using the digits from 1 to 9, each at most once, make two 2-digit numbers whose product is as small as possible.

$$\begin{array}{c} \square \square \times \square \square \\ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \end{array}$$

EXPLORATION: Can you apply what you learned for multiplying two numbers to do this with multiplying three two-digit numbers? Can you think of other interesting variations?

Puzzle of the Week

Combining Digits – Easy as 1 2 3 4

Here are some ways to get 0 and 1 using 1, 2, 3, and 4.

$$0 = 1 + 4 - 2 - 3$$

$$0 = (3 - 1 - 2) * 4$$

$$1 = (2 - 1) * (4 - 3)$$

$$1 = 4 - 3 * (2 - 1)$$

THE CHALLENGE: How many numbers can you get using each of the numbers 1, 2, 3, and 4 in any order, using addition, subtraction, and multiplication?

EXPLORATION: How many more numbers can you make if you are also allowed to make two-digit numbers with the digits? For example, $26 = 24 + 3 - 1$.

Puzzle of the Week

Fractions – 4

THE CHALLENGE: Use the numbers 1 to 9 at most once each to fill the boxes in each of these puzzles. Make these two-digit fractions as close to the target numbers from 1 to 5 as possible, without equalling the number.

For example, $85/17 = 5$ is not allowed as a contender, but $83/17$ is okay and is somewhat close to 5.

$$\frac{\square\square}{\square\square} \sim 1$$

$$\frac{\square\square}{\square\square} \sim 2$$

$$\frac{\square\square}{\square\square} \sim 3$$

$$\frac{\square\square}{\square\square} \sim 4$$

$$\frac{\square\square}{\square\square} \sim 5$$

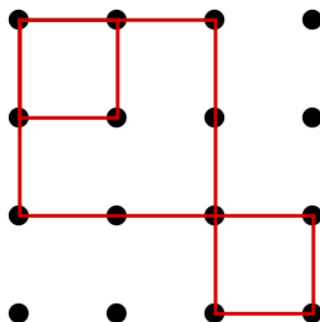
1 2 3 4 5 6 7 8 9

EXPLORATION: Look at what happens for targets from 6 to 10. What changes in your strategy for targets above some value?

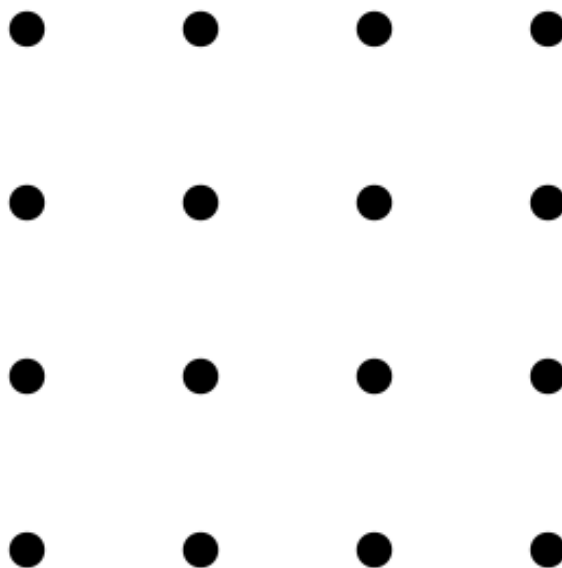
Puzzle of the Week

Finding Squares – 1

Drawn in red in this grid are two 1 by 1 squares and one 2 by 2 square with horizontal and vertical sides.



THE CHALLENGE: Find the number of squares of all sizes with horizontal and vertical sides in this grid.



EXPLORATION: Play with even bigger grids. Can you discover a systematic way to count all the squares? How does your answer change, if at all, if the grid of points is a rectangle that is not a square?

Puzzle of the Week

Moving Digits – 2

THE CHALLENGE: Find a 4-digit number, ABCD, that satisfies this interesting equation when you reverse the digits:

$$ABCD \times 4 = DCBA$$

EXPLORATION: Investigate why this cannot happen for numbers less than 1000. Also, look for numbers larger than 9,999 that have this property.



Puzzle of the Week

Combining Digits – Five 2's

Here are some ways to get 0 and 1 using five 2's.

$$0 = (22 - 22) \times 2$$

$$0 = (2/2 - 2/2) \times 2$$

$$1 = 2 - (2/2 \times 2/2)$$

$$1 = (2/2 + 2/2) / 2$$

THE CHALLENGE: How many numbers can you get using five 2's using addition, subtraction, multiplication, division and creating double-digit numbers?

2 2 2 2 2

EXPLORATION: Play with other groups of identical numbers. Are some more interesting than others?

Puzzle of the Week

Fractions – 6

THE CHALLENGE: Use the numbers 1 to 9 at most once each to fill in these boxes. How many solutions can you find?

$$\frac{\square}{\square} + \frac{\square}{\square} = \square$$

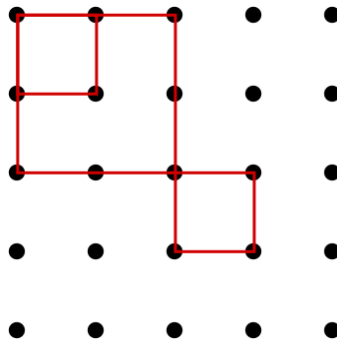
1 2 3 4 5 6 7 8 9

EXPLORATION: Are there any single-digit values for the right side of the equation that are impossible?

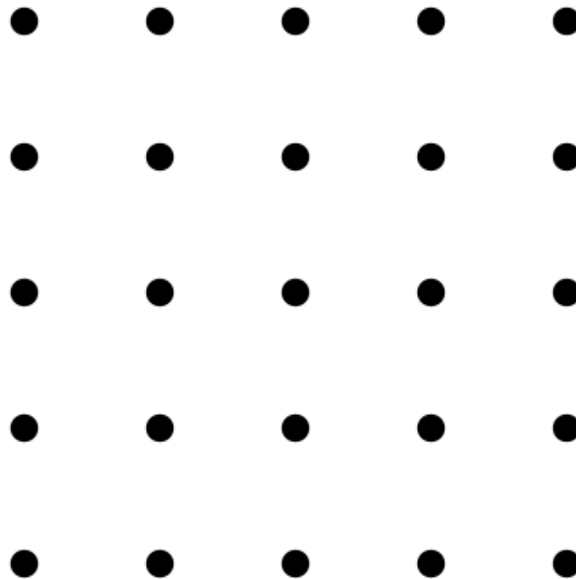
Puzzle of the Week

Finding Squares – 2

Drawn in red in this grid are two 1 by 1 squares and one 2 by 2 square with horizontal and vertical sides.



THE CHALLENGE: Find the number of squares of all sizes and orientations in this grid. Unlike what is shown in the introduction, not all of these squares will have horizontal and vertical sides!



EXPLORATION: Play with what happens with larger grids and with grids that are not squares.

Puzzle of the Week

Water Cups – 1

You have two unmarked water cups. One holds 3 ounces and the other 7 ounces. You also have a large supply of water. You can use these two cups to create amounts other than 3 ounces and 7 ounces. For example, create 4 ounces in the larger cup by filling the 7-ounce cup and then pouring 3 of its ounces into the smaller cup.

THE CHALLENGE: Describe the steps for putting 2 ounces in one of these cups.



3 Ounce



7 Ounce

EXPLORATION: Describe the steps to take to create any amount from 1 to 7 ounces. What is your general method? Experiment with what happens if your pair of cups have different sizes. For example, what happens if the sizes in ounces are 4 and 7, or 5 and 11?

Puzzle of the Week

Turning the Tables

THE CHALLENGE: This started out as a standard multiplication table for the numbers 2 through 9. Then the rows and columns were all mixed up. Finally, most of the numbers have been removed. Put in all the missing numbers!

X								
2								
		40						
				49				
	20					36		
		72						
			9					12
					48			

EXPLORATION: Make one of these puzzles for someone else. How many numbers can you leave out and still have a puzzle that can be completely solved?

Puzzle of the Week

Fractions – 7

THE CHALLENGE: Use the numbers 1 to 9 at most once each to fill in these boxes. Do not use improper fractions. How many solutions can you find?

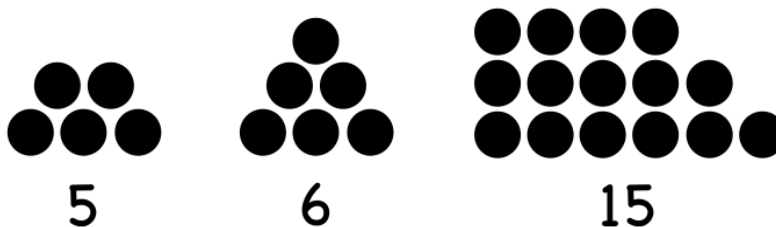
$$\frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square}$$

1 2 3 4 5 6 7 8 9

Puzzle of the Week

Trapezoidal Numbers – 1

Trapezoidal Numbers are the sum of two or more consecutive numbers. They deserve their name because you can make a trapezoid with that many dots, as pictured in the examples below. Note that having 1 dot on the top row is stretching the idea of being a trapezoid a bit, but it is allowed for these numbers.



THE CHALLENGE: Show that every odd number is a Trapezoidal Number.

Odd

=



EXPLORATION: Can you find some even numbers that are Trapezoidal Numbers? Can you find some that aren't?

Puzzle of the Week

Water Cups – 2

You have two unmarked water cups. One holds 9 ounces and the other 15 ounces. You also have a large supply of water. You can use these two cups to create amounts other than 9 ounces and 15 ounces. For example, create 6 ounces in the larger cup by filling the 15-ounce cup and then pouring 9 of its ounces into the smaller cup.

THE CHALLENGE: Find all the amounts that you can create using these two cups.



9 Ounce



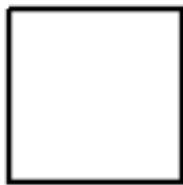
15 Ounce

EXPLORATION: Investigate other pairs of water cups that involve two numbers with a common divisor greater than 1. What patterns do you notice?

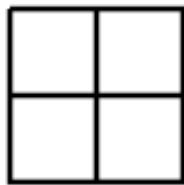
Puzzle of the Week

Filling Squares with Squares

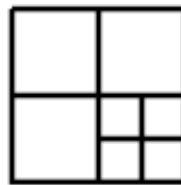
Here is how to fill one large square with 1, 4, or 7 squares.



1

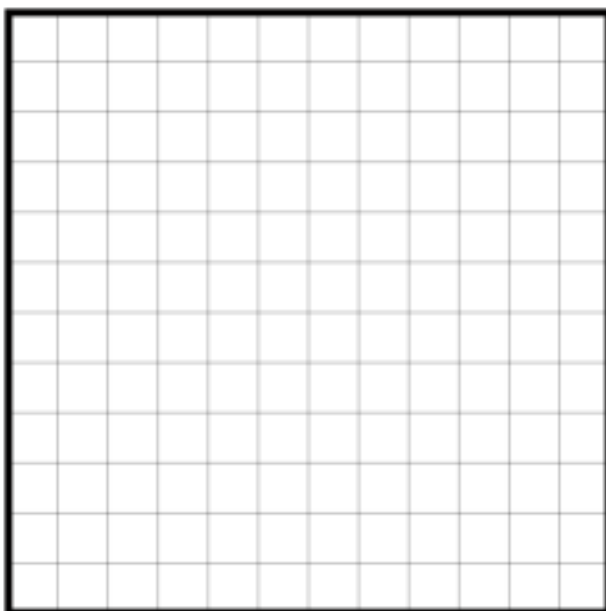


4



7

THE CHALLENGE: Find other square counts for filling a large square. Can you do it for 2, 3, 5, 6, 8, 9, or 10 squares?



EXPLORATION: When possible, find more than one way to get some of these numbers.

Puzzle of the Week

Fractions – 9

THE CHALLENGE: Use the numbers 2 to 9 at most once each to fill in these boxes. There are a lot of answers that are essentially the same, so organize them with increasing numerators and denominators.

$$\frac{\square}{\square} \times \frac{\square}{\square} = 1$$

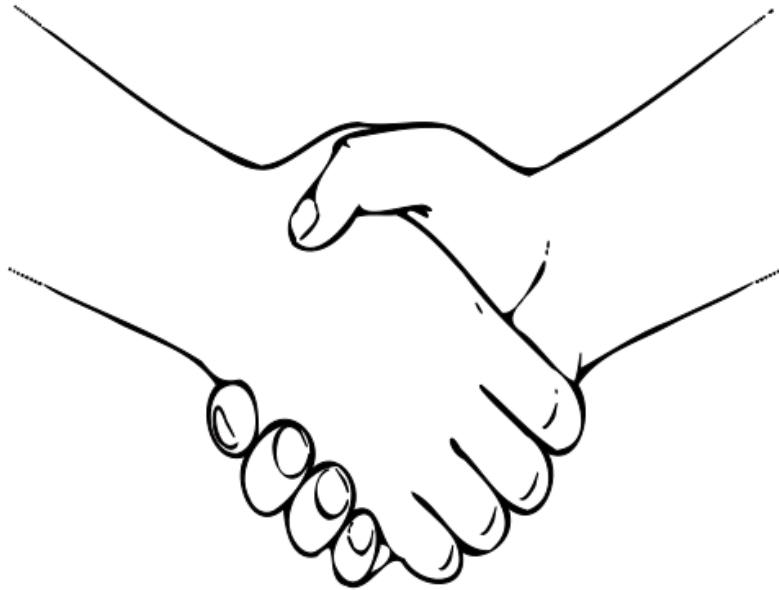
2 3 4 5 6 7 8 9

Puzzle of the Week

Handshakes at a Party – 2

Eight people were at a party. A lot of handshaking took place. When asked how many hands each person shook, they were amazed to discover that each number was different. When this was announced to the group, one person yelled out “That’s impossible!”

THE CHALLENGE: Was that person right, did someone make a mistake in their handshake count? How do you know?



Puzzle of the Week

Pirates with Gold – 1

There are three pirates on an island, and they have a total of 12 gold coins.

The rules are:

1. The pirates are very smart.
2. Each pirate wants as much of the gold as possible for themselves, and does not care about the others.
3. The youngest pirate must propose a plan for splitting up the gold. If the plan is acceptable to over half of all the pirates (including the youngest), then the plan is adopted. Otherwise, the youngest pirate is forced to leave the island with no gold, and the new youngest pirate must propose a plan.

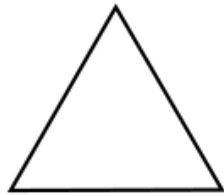
THE CHALLENGE: What is the most gold coins the youngest pirate can get in an acceptable plan?



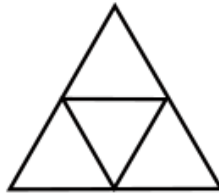
Puzzle of the Week

Filling Triangles with Triangles

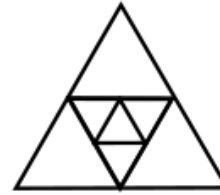
Here's how to fill one large triangle with 1, 4, or 7 triangles..



1

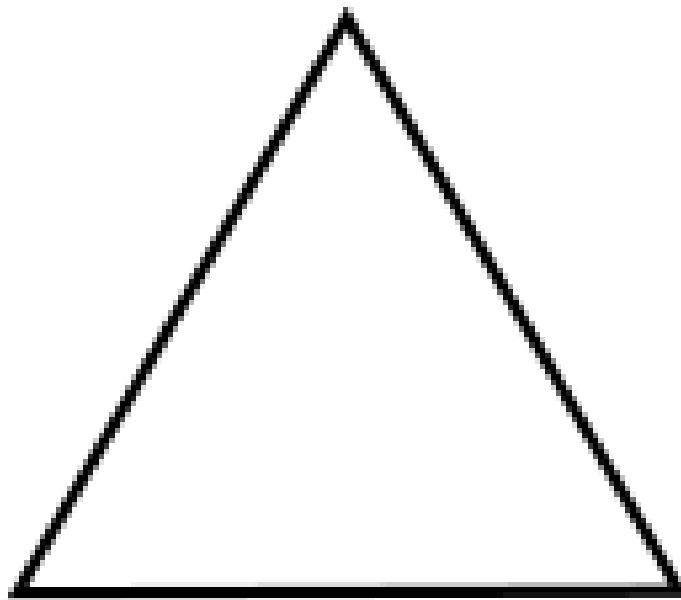


4



7

THE CHALLENGE: Find other triangle counts for filling a large triangle. Can you do it for 2, 3, 5, 6, 8, 9, or 10 triangles?



EXPLORATION: When possible, find more than one way to get some of these numbers.

Puzzle of the Week

Fractions – 12

THE CHALLENGE: Use the numbers 1 to 9 at most once each in each set of boxes. First, make

$\frac{\square}{\square} \times \frac{\square}{\square}$ equal to $2/3$, and then find values that make $\frac{\square}{\square} \times \frac{\square}{\square}$ as close as possible to $5/11$.

$$\frac{\square}{\square} \times \frac{\square}{\square} = \frac{2}{3}$$

$$\frac{\square}{\square} \times \frac{\square}{\square} \sim \frac{5}{11}$$

1 2 3 4 5 6 7 8 9